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2. Find the dimension of the nullspace

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Recitation due Jan 24, 2024 20:30 IST

Find the dimension of the nullspace (External resource)

(1.0 points possible)

Find the dimension of the nullspace using null()

In this problem, we will find the nullspace and the dimension of the matrix defined below

```
A = [0.9653    0.0345    0.8963   -0.8830   -0.4327    0.9822    1.381
      -0.4002   -0.3695    0.3388   -0.7290   -0.0606   -1.3775    0.637
        0.6734   -0.2631    1.1996    0.6106    0.1278    0.7653    0.453
        0.9755   -0.2065    1.3885    0.4758    0.5416    0.1616    1.247
        0.2546   -0.4360    1.1266    0.8531   -0.7262    2.1241   -0.888
        0.1431   -0.8431    1.8293   -0.7305   -0.5756   -0.2793    1.141
       -0.2426    0.3752   -0.9930    0.9713   -0.0005    1.1049   -1.585
        0.5974   -0.6121    1.8216    0.0908   -0.5123    1.1007    0.606
```

Create a matrix **N** whose columns form a basis for the nullspace of **A** by checking the command

```
null( )
```

and create a variable

```
dim
```

whose value is the dimension of the nullspace.

Finally, create a new matrix **B** = **A****N**. Is the matrix **B** a matrix of zeroes? Why or why not? Discuss this numerically in the discussion forum.

Script ?

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 [MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```
1 %Copy and past the matrix A (using cmd-c cmd-v on mac and ctrl-c ct
2 %and find the dimension of its nullspace
3 %You must create a matrix N and variable dim as described above.
4 A = [0.9653    0.0345    0.8963   -0.8830   -0.4327    0.9822    1.
5      -0.4002   -0.3695    0.3388   -0.7290   -0.0606   -1.3775    0.
6        0.6734   -0.2631    1.1996    0.6106    0.1278    0.7653    0.
7        0.9755   -0.2065    1.3885    0.4758    0.5416    0.1616    1.
8        0.2546   -0.4360    1.1266    0.8531   -0.7262    2.1241   -0.
9        0.1431   -0.8431    1.8293   -0.7305   -0.5756   -0.2793    1.
10       -0.2426    0.3752   -0.9930    0.9713   -0.0005    1.1049   -1.
11        0.5974   -0.6121    1.8216    0.0908   -0.5123    1.1007    0.
12
13 N = null( A );
14 dim = size(N, 2);
15 %Finally, look at B = A*N. Is this a matrix of zeroes? Why or why r
16 %Note that numerically, values close to zero should be assumed to b
17 %But why is it not an exact equality?
18
19 B = A*N
```

 Run Script 

2. Find the dimension of the nullspace

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